Database Management Systems

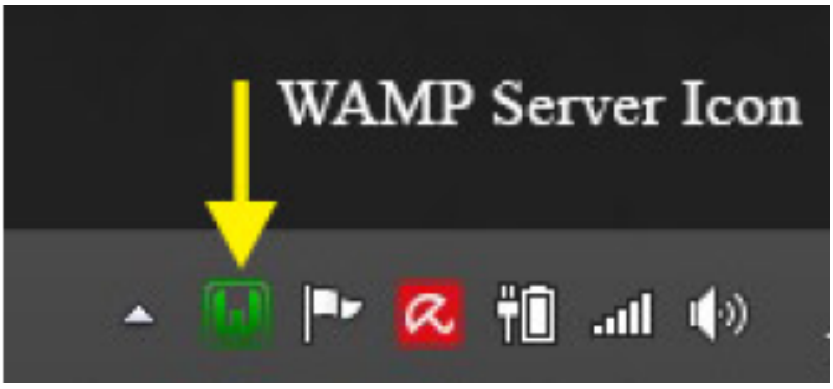
Lab-1: Introduction to MySQL and Data Definition Languages(DDL)

What is the difference between wampserver and mysqlserver?
WAMP server is a bundle of different programs that are used together: Apache (A Web server), mySQL (A database server), and PHP (A programming language).
MySQL is an open-source Relational Database Management System (RDBMS). To set up your environment, install WampServer. It includes the MySQL database we will use for this lab.

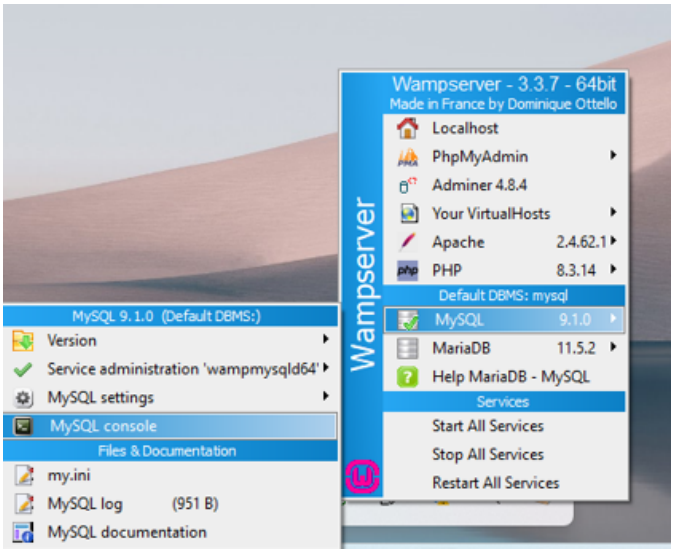
Installation of WampServer

- Make sure to first download and install Visual C++ Redistributable Packages available from here
- After installing the required Visual C++ Redistributable Packages, download WampServer From the same page here or from WampServer
- After installing Wamp Server start the Wamp serve:

  1. Click the Wampserver shortcut from your Desktop
  2. Go to on right bottom corner of your Desktop and press the hidden icons
  3. To know that whether the wampserve is running or not, look in the system tray for the Wampserver icon as shown in the below screenshot. If color is green, then the server is running. If its color is red, then one or more services had failed to start.



  4. Press the WampServer icon and select MySQL



5. Select MySQL Console
6. When you install MySQl server you are given a username and password by default
   - Username: root
   - Password: (no password)

```
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 46
Server version: 9.1.0 MySQL Community Server - GPL

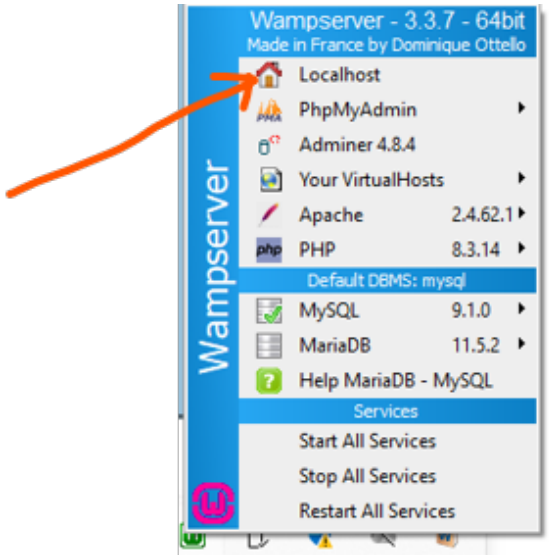
Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

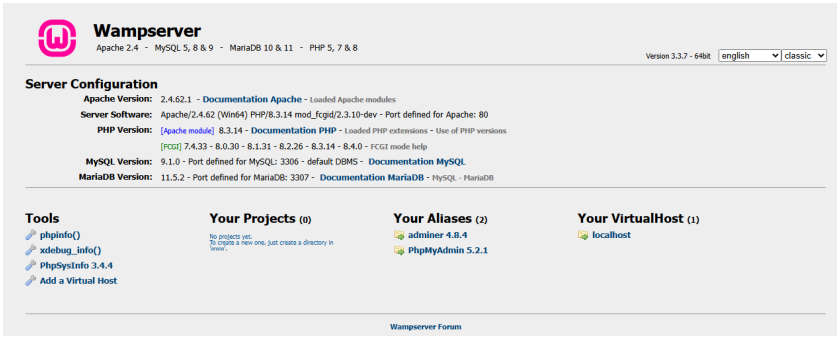
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

7. To test whether the server is running or not in the browser click the wampserver icon and select the Localhost option as shown in the below screenshot:



8. An alternative to test manually is, open the browser and in the address bar type the URL `http://localhost` and hit Enter Key.
9. If everything is fine(Wampserver is running smoothly), in the browser you will see the page as shown in the below screenshot after completing the above step.



That's it! you have installed Wampserver successfully on your machine.

Introduction to SQL

Objectives:

1. Implementation of DDL commands of SQL with suitable examples
    - Create table
    - Alter table

- Drop Table

2. Implementation of "Insert" DML commands of SQL with suitable examples

SQL (Structured Query Language) is a database computer language designed for managing data in relational database management systems (RDBMS).

In RDBMS:

- Tables are the basic building blocks of a database.
- Columns define the categories of information in the table
- Rows represent individual records in the table.

SQL is provided in two modes.

- Interactive SQL: This is mode is used to operate directly on a database that is the response to any SQL command can be seen almost immediately on the same terminal.
- Embedded SQL: Embedded SQL consists of SQL commands used within programs written in some other language like C, C++, Java, Python etc...

Creating and Accessing a Database
To create a new database in SQL, we use CREATE DATABASE command followed by the database name. Database names cannot contain spaces; if needed, an underscore (_) can be used instead.
Syntax:

```
CREATE DATABASE database_name;
```

Let's create a database called StudentInfo:

```
CREATE DATABASE StudentInfo;
```

✓ Use the show statement to find out what databases currently exist on the server:

- mysql>show databases;

✓ It will display all list of database.

- If the test database exists, try to connect/access it:
  - mysql>use Database Name

Deleting a Database
If you ever want to delete a database, use:

```
DROP DATABASE Database_name;
```

Creating a Table
In relational databases, a table is a structured set of data organized into rows and columns:

- Rows represent records.
- Columns represent attributes of data.

To create a new table, you use the CREATE TABLE statement. Here's the basic syntax of the CREATE TABLE statement

```
CREATE TABLE table_name (
column1 datatype constraint,
column2 datatype constraint,
...
);
```

In this syntax:

- table_name is the name of the table you want to create.
- column1, column2, ... are the column names of the table.
- Each table column is associated with a data type it can store and an optional rule for data. In the database, we call this rule a constraint.

Data Types in MySQL
String Data Types

```
CHAR(size), VARCHAR(size), BINARY(size),
    BLOB(size), ENUM(val1, val2, val3, ...),
    LONGBLOB, LONGTEXT, MEDIUMBLOB, MEDIUMTEXT,
    SET, TEXT(size), TINYTEXT, VARBINARY(size),
```

Numeric Data Types

```
BIGINT (size), BIT (size), BOOLEAN, DECIMAL (M, D) or
    NUMERIC (M, D), DOUBLE (size, d) or REAL (size,
    d), FLOAT (size, d), INT (size) or INTEGER (size),
    MEDIUMINT (size), SMALLINT (size), TINYINT (size)
```

Date and Time Data Types

```
DATE: YYYY-MM-DD, DATETIME(fsp): YYYY-MM-DD
    HH:MM:SS, TIMESTAMP(fsp), YEAR,
```

Constraints: The following are the most common constraints used in MySQL

```
NOT NULL, CHECK, DEFAULT, PRIMARY KEY,
    AUTO_INCREMENT, UNIQUE, UNIQUE, FOREIGN KEY,
    DEFAULT
```

The following example creates a table called ”Students” that contains six columns: StudentID, FirstName, MiddleName, LastName, BirthDate, Nationality.

```
CREATE TABLE Students (
ID INT NOT NULL,
FirstName VARCHAR(255) NOT NULL,
MiddleName VARCHAR(255) NOT NULL,
LastName VARCHAR(255) NOT NULL,
BirthDate DATE NOT NULL,
Nationality CHAR(50),
UNIQUE (ID)
);
```

Alter Table:
SQL ALTER TABLE statement modify the structure of an existing table in a database. Whether adding new columns, modifying existing ones, deleting columns or renaming them. ALTER TABLE statement enables you to make changes without losing data stored in the table.
Syntax:

```
ALTER TABLE table_name [ADD | DROP | MODIFY] column_name
    datatype;
```

- table_name: name of the table you want to modify.
- ADD: used to add a new column.
- DROP: used to remove an existing column.
- MODIFY: used to change datatype or definition of an existing column.

Examples:
To create and add a new column called ”City” in the existing ”Student” table:

```
ALTER TABLE Students
ADD City VARCHAR(30);
```

To alter the data type of the ”FirstName” column to CHAR with a size of 50:

```
ALTER TABLE Students
MODIFY COLUMN FirstName CHAR(50);
```

To drop the ”City” column from the ”Student” table:

```
ALTER TABLE Students
DROP COLUMN City;
```

RENAME COLUMN : enables to change a column’s name while maintaining its data type and contents.

```
ALTER TABLE Students
RENAME COLUMN FirstName TO GivenName;
```

You can rename an entire table using the RENAME TO clause. This changes the name of the table while preserving its structure and data.

```
ALTER TABLE Students
RENAME TO Employee;
```

Drop Table
The DROP TABLE command in SQL permanently removes a table from the database, including its data and structure.
Syntax:

```
DROP TABLE table_name;
```

Insert:
INSERT INTO statement is used to add new records into a table.

Syntax:

```
INSERT INTO table_name
VALUES (value1, value2, value3, ...);
```

Let’s insert values to the table ”Students” we have created:

```
INSERT INTO Students (ID, FirstName, MiddleName,
    LastName, BirthDate, Nationality, City) values
    (1, 'Yabsra', 'Daniel', 'Alem', 20100723,
    'Ethiopia', 'Mekelle');
```

Inserting Multiple Rows at Once: You can insert multiple rows into a table in a single query.

```
INSERT INTO Students (ID, FirstName, MiddleName, LastName,
    BirthDate, Nationality, City) values
(2, 'Zaid', 'Ataklti', 'Gidey', 20050125, 'Ethiopia', 'Hawassa'),
(3, 'Kamau', 'Karanja', 'Njoki', 19951208, 'Kenya', 'Nairobi'),
(4, 'Ramses', 'Anubis', 'Osiris', 20100115, 'Egypt', 'Cairo'),
(5, 'Marco', 'Pollini', 'Andrea', 20071112, 'Italy', 'Florence');
```

Display the table schema using the DESC command and view its records with a SELECT statement.

```
mysql> desc Students;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| ID          | int          | NO   | PRI | NULL    |       |
| FirstName   | varchar(255) | NO   |     | NULL    |       |
| MiddleName  | varchar(255) | NO   |     | NULL    |       |
| LastName    | varchar(255) | NO   |     | NULL    |       |
| BirthDate   | date         | NO   |     | NULL    |       |
| Nationality | char(50)     | YES  |     | NULL    |       |
| City        | char(30)     | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
```

```
mysql> select * from Students;
+----+-----------+------------+----------+------------+-------------+----------+
| ID | FirstName | MiddleName | LastName | BirthDate  | Nationality | City     |
+----+-----------+------------+----------+------------+-------------+----------+
|  1 | Yabsra    | Daniel     | Alem     | 2010-07-23 | Ethiopia    | Mekelle  |
|  2 | Zaid      | Ataklti    | Gidey    | 2005-01-25 | Ethiopia    | Hawassa  |
|  3 | Kamau     | Karanja    | Njoki    | 1995-12-08 | Kenya       | Nairobi  |
|  4 | Ramses    | Anubis     | Osiris   | 2010-01-15 | Egypt       | Cairo    |
|  5 | Marco     | Pollini    | Andrea   | 2007-11-12 | Italy       | Florence |
+----+-----------+------------+----------+------------+-------------+----------+
```